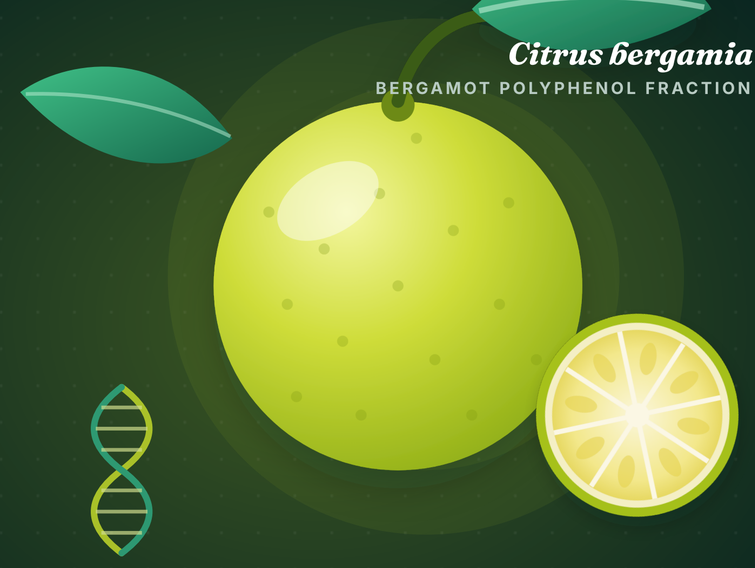
Al Ain University
COLLEGE OF PHARMACY
GENETICS · PREVENTION
NUTRACEUTICALS



Before the *Statin*

Detecting genetic and early cardiometabolic risk at 15–18, and testing *Citrus bergamot* as a first step

Noor Al Ahmad • Al Ain University, UAE • Evidence review & proposed RCT

−54.5% — **CHD per 1 mmol/L lower LDL-C, lifelong**
≈3× the benefit of lowering it later · Ference, JACC 2012

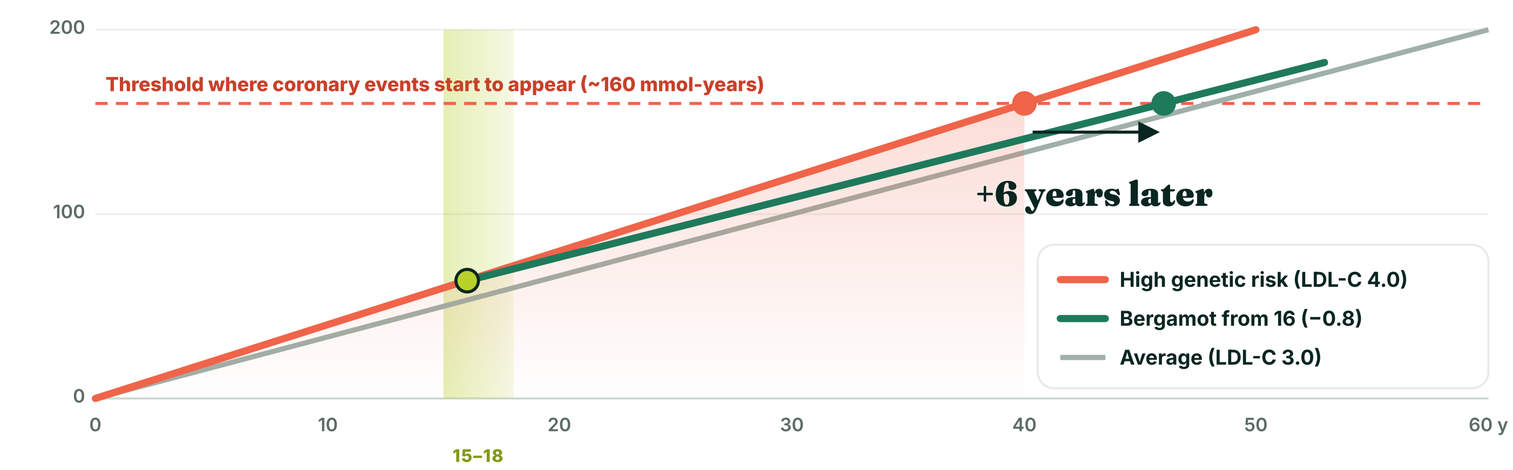
≈46% — **lower risk with a healthy lifestyle**
even in people with high genetic risk · Khera, NEJM 2016

−40 mg/dL — **LDL-C with bergamot**
pooled adult RCTs, low certainty · Chambari, Food Sci Nutr 2026

01 Risk starts in childhood, *not at 40*

LDL-C damage is cumulative: every year of exposure counts.

CUMULATIVE LDL-C EXPOSURE BY AGE (ILLUSTRATIVE MODEL)



Conceptual curves (LDL-C × years). Each line assumes a constant LDL-C; the earlier the slope falls, the later the threshold is reached. Ference 2012 & 2017; Jacobs (i3C), NEJM 2022.

Childhood risk → adult events
Risk factors at 3–19 y predict adult CV events (38,589 followed)

Genes are not destiny
High polygenic risk, healthy lifestyle: ≈46% fewer coronary events

A screening window exists
Universal lipid screening at 9–11 and 17–21 y (NHLBI)

Most teens don't meet statin criteria
A gap between "advice only" and drug therapy

02 Detect

Five checks that flag a high-risk teenager early.

GENETIC & EARLY RISK SCREEN · AGE 15–18

Polygenic risk score
CAD PRS from a genotyping array

Lp(a), once
Genetically set; measure at least once

Family history
Premature CVD in a 1st-degree relative

Lipids & apoB
Non-fasting LDL-C, TG, HDL-C

Metabolic profile
Waist, blood pressure, glucose / HOMA-IR, liver fat (MASLD), BMI

PATHWAY A · PROPOSED

Polygenic or metabolic risk
LDL-C below drug thresholds, no monogenic variant → **lifestyle + bergamot**, re-check every 6–12 months

PATHWAY B · GUIDELINE CARE

Monogenic FH / very high LDL
LDLR / APOB / PCSK9 variant → **statin as indicated**. Bergamot is not a substitute here.

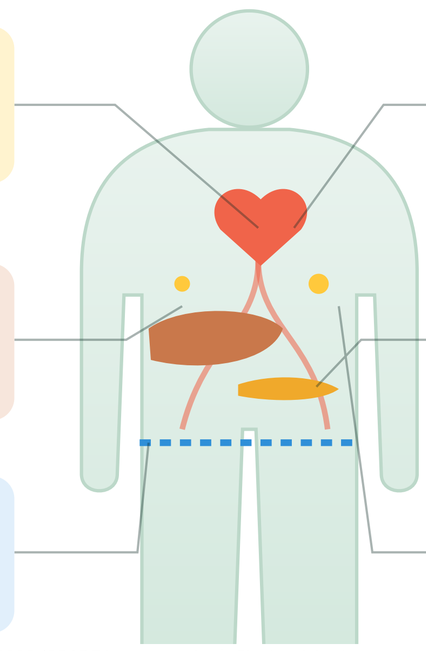
03 Treat *early*

One supplement, Citrus bergamot, acting on several linked risk factors.

POOLED EFFECTS IN ADULT RCTS (9 TRIALS, 615 ADULTS) AND SINGLE TRIALS

BLOOD LIPIDS
−40 mg/dL LDL-C
TG −52 · HDL-C +5.6 mg/dL

BLOOD PRESSURE
−2.1 mmHg systolic
pooled, 9 RCTs



FATTY LIVER (MASLD)
−48 dB/m liver fat
vs −27 placebo · Ferro 2020

GLUCOSE & INSULIN
−12 mg/dL glucose
HOMA-IR −0.38 · pooled

WAIST & VISCERAL FAT
−1 cm waist (pooled)
visceral fat ↓ · Rondanelli 2020

VESSELS & INFLAMMATION
hs-CRP ↓
endothelial function ↑ · Fogacci 2022

Pooled values: Chambari et al., Food Sci Nutr 2026 (GRADE low to very low). Single-trial values are from adults; several used combination products.

04 The evidence so far

Randomised trials of bergamot (adults), plus the closest trial in young people.

Gliozzi 2013 · n = 77
Mixed hyperlipidaemia · BPF 1000 mg ± rosuvastatin, 30 d
↓LDL-C alone, and boosted the statin's effect

Mollace 2019 · n = 60
Type 2 diabetes + mixed hyperlipaemia · double-blind
↓LDL ↓TG ↓glucose, ↑HDL, fewer small dense LDL

Capomolla 2019 · n = 52
Metabolic syndrome, obesity · 90 days, 2 doses
−18% glucose · TG −32% · weight ↓

Rondanelli 2020 · n = 64
Overweight, mild ↑cholesterol · 12 weeks
↓VAT visceral fat, TC and LDL-C vs placebo at 30 d

Ferro 2020 · n = 102
Fatty liver · BPF + wild cardoon, 12 weeks
↓liver fat −48 vs −27 dB/m (placebo)

Fogacci 2022 · n = 60
Polygenic hypercholesterolaemia · combination, 8 weeks
↓LDL-C ↓hs-CRP, ↓fatty liver index vs placebo

Stonehouse 2025 · n = 42
Mild ↑LDL-C · sterols + bergamot + artichoke, 4 months
No effect on LDL-C vs placebo (−0.12 mmol/L, NS)

Guardamagna 2011 · age 8–16
Children with FH / FCH · red yeast rice + policosanols
−25% LDL-C; well tolerated (not bergamot)

−55 mg/dL — **LDL-C · 14 trials**
Sadeghi-Dehsahraei, Phytother Res 2022

−40 mg/dL — **LDL-C · 9 RCTs, 615 adults**
Chambari, Food Sci Nutr 2026 · GRADE low–very low

#1 — **of 10 nutraceuticals for LDL-C**
Network MA, 131 trials · Osadnik, Pharmacol Res 2022

The gap
No bergamot RCT in anyone under 18 has been published. That is what this study would test.

PROPOSED STUDY

Genetic screening + *Citrus bergamot* at 15–18: fewer risk factors before statins are needed?

1. **Screen** — Ages 15–18: PRS, Lp(a), family history, lipids, metabolic profile
2. **Select** — High polygenic or metabolic risk; monogenic FH referred to guideline care
3. **Randomise** — Bergamot polyphenol fraction vs placebo, both with lifestyle advice
4. **12 months** — Double-blind, with adherence and safety checks (liver, muscle, growth)
5. **Outcomes** — LDL-C / apoB, TG, HOMA-IR, BP, waist, liver fat, carotid IMT

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Safety note: bergamot is a supplement, not an approved therapy. Teens with monogenic FH or very high LDL-C follow guideline statin care.

KEY SOURCES

Ference JACC 2012 · EHJ 2017
Khera NEJM 2016
Jacobs (i3C) NEJM 2022
Chambari Food Sci Nutr 2026
Sadeghi-Dehsahraei Phytother Res 2022
Osadnik Pharmacol Res 2022
NHLBI Pediatrics 2011 · **EAS Lp(a)** 2022
+ 8 randomised trials (section 04)

Scan for all 18 references
Full citations with clickable DOI links